

FULL MARKING SCHEME

IS2200

IS Analysis and Design for Business

Winter 2025 Examination · Paper 1

Module	IS2200 — IS Analysis and Design for Business
Session	Winter 2025
Duration	1.5 hours
Structure	Section A (compulsory) · one question from Section B
Total marks	100 (50 + 50)

SECTION A — COMPULSORY

Question 1

- (a) Create a use case diagram outlining the identified users, and how they interact with the system. **(10 marks)**
Clearly label the users, use cases, and associations.

FULL MODEL ANSWER

Scope note. Question 1 is based on a single extended scenario (Gym+Coffee's online order-fulfilment system) spanning parts (a)–(c). The worked answers below build one consistent model of the system across all three parts, so entities, processes and data stores introduced in (a) and (b) are reused — not re-invented — in (c), exactly as an examiner would expect for a scenario that is developed part-by-part.

The use case diagram models the Gym+Coffee order-fulfilment system as a single system boundary, with the external users (actors) who interact with it shown outside that boundary, and the discrete pieces of system functionality they trigger shown as ellipses (use cases) inside it. Reading the scenario line by line surfaces four human/role actors and one external system actor:

- **Customer** — creates/updates an account, adds items to the cart, checks out (submitting payment), and tracks the order.
- **Administrator** — monitors order progress, resolves exceptions (failed authorisation, address mismatch, out-of-stock item), and generates the packing slip / books the courier.
- **Warehouse Staff** — picks and packs the order once instructed.
- **Courier** — collects the parcel and updates delivery tracking events.
- **Payment Gateway** — an external system (not a human user) that authorises the payment transaction; modelled as an actor because it is outside the system boundary but exchanges information directly with it.

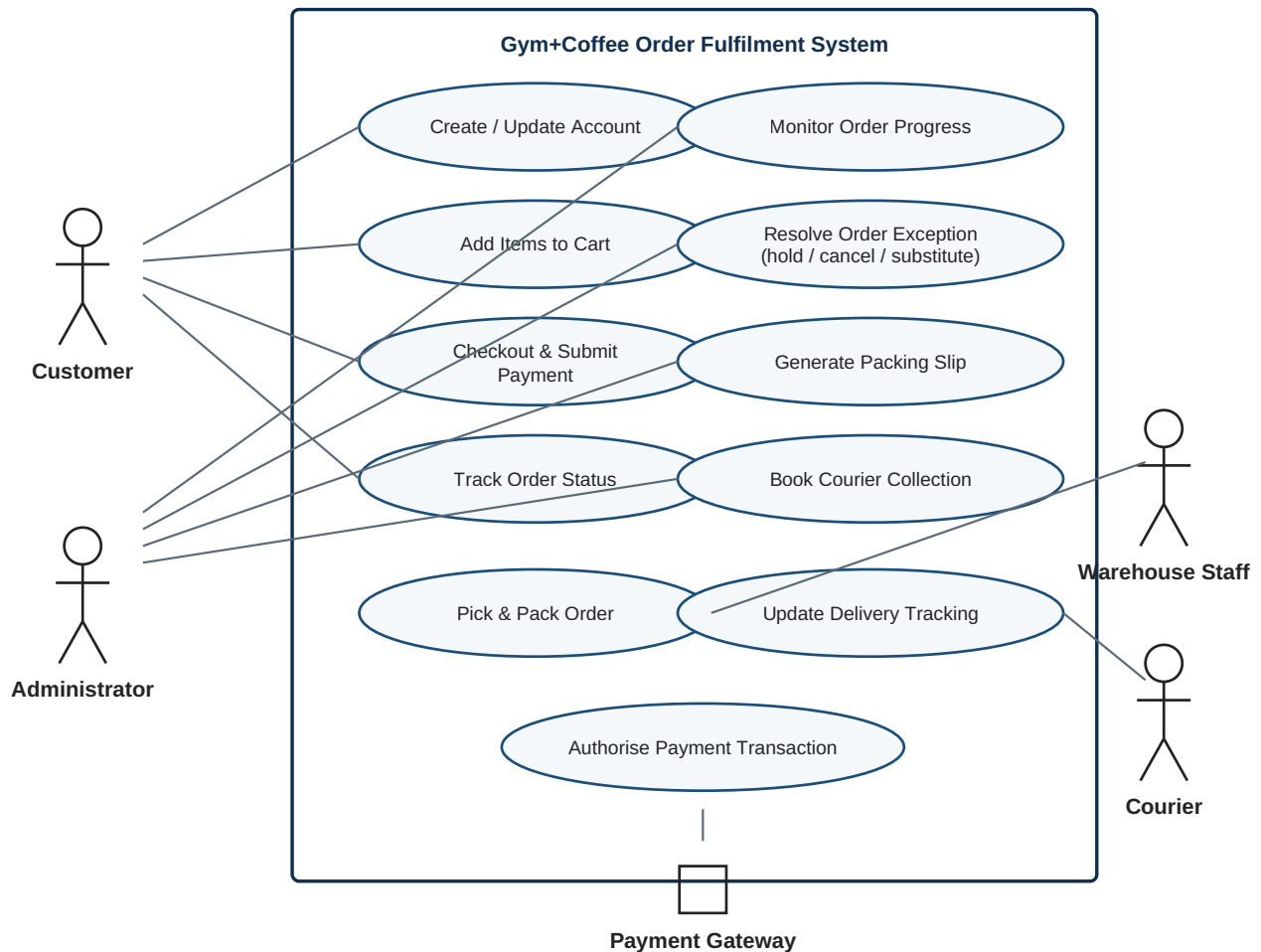


Figure 1(a). Use case diagram for the Gym+Coffee order-fulfilment system, showing four human actors, one external system actor, and their associations to the ten identified use cases.

Each association line is drawn as a plain (unlabelled, undirected) line between actor and use case, per standard UML convention — direction and message content are not shown on a use case diagram, only participation. "Resolve Order Exception" is drawn as a single use case rather than three separate ones (hold / cancel / substitute) because the scenario treats these as alternative actions within one administrator task, not three independent goals a user initiates.

Mark Allocation — (a)

Component	Marks	Criteria for full marks
All four human actors + external system actor correctly identified	3	Customer, Administrator, Warehouse Staff, Courier, and Payment Gateway all present and correctly labelled.
Use cases correctly identified and meaningfully named	4	Use cases reflect discrete goals from the scenario (not implementation steps); named as verb phrases; no missing major function.
Associations correctly drawn between actors and use cases	2	Every use case connected to at least one actor; no actor left floating; no association drawn to an unrelated use case.
Correct UML notation	1	System boundary box, ellipse use cases, stick-figure/actor notation all used correctly and consistently.
Total	10	—

(b) Create a context level data flow diagram to provide an overview of the system's interactions with **(15 marks)** external entities. Label all elements appropriately.

FULL MODEL ANSWER

A context-level (Level 0) DFD shows the system as a single process bubble, with every external entity that sends or receives data across the system boundary, and the major data flows between them. No internal processes and no data stores are shown at this level — that detail is reserved for the decomposition in part (c).

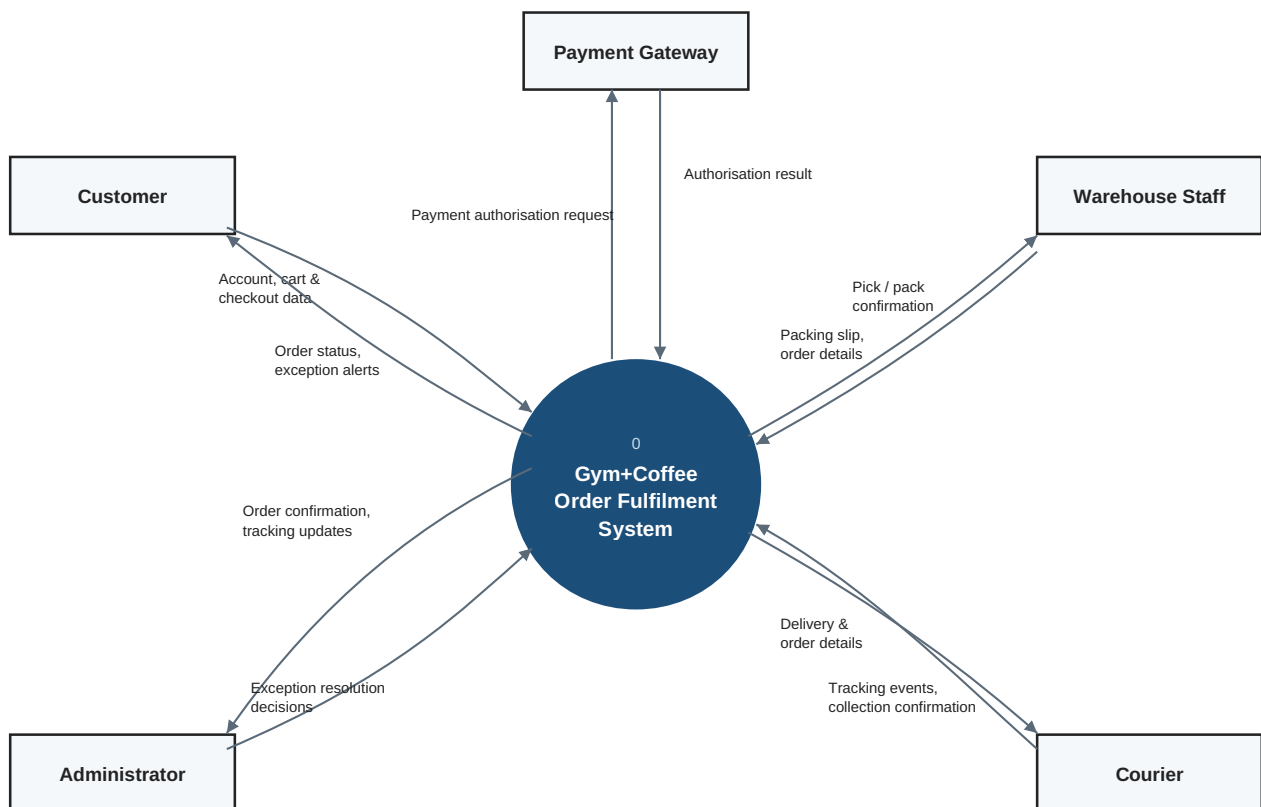


Figure 1(b). Context-level DFD: the Gym+Coffee order-fulfilment system as a single process (0), bounded by five external entities and their data flows.

The five external entities carried over from part (a) are retained here, since a context diagram's entities are exactly the actors who cross the system boundary in a use case diagram — the two diagrams should agree on who the external parties are, even though they show different things about the system. Data flows are drawn as directional, labelled arrows: *account/cart/checkout data* and *order confirmation/tracking updates* between Customer and the system; *exception decisions* and *status/alerts* between Administrator and the system; *packing slip/order details* out to Warehouse Staff and *pick/pack confirmation* back; *delivery details* out to Courier and *tracking events* back; and a paired *authorisation request/result* flow with the Payment Gateway.

The exam paper labels part (b) "context level" and part (c) a separate "level 0" diagram that expands it. The module's own lecture slides use "Context Level (Level 0)" as a single combined term for what part (b) alone is asking for, then call the next decomposition step "Level 1." This answer follows the exam paper's own wording literally: part (b) = the single-process context diagram; part (c) = the first decomposition into numbered internal processes (i.e. what the lecture slides label "Level 1"). Markers using the lecture's own numbering should read "Level 0" in part (c) of the paper as referring to that same first-decomposition diagram.

Mark Allocation — (b)

Component	Marks	Criteria for full marks
Single process correctly represents the whole system	3	Exactly one process, numbered 0, labelled with the system name; no internal detail shown.
All five external entities correctly identified	5	1 mark each: Customer, Administrator, Warehouse Staff, Courier, Payment Gateway all present and correctly drawn as external entities (not processes or stores).
Data flows correctly labelled and directional	5	Arrows show a sensible, correctly directed flow of named data for every entity (at least one inbound, one outbound where the scenario supports it).
Correct DFD notation used throughout	2	Circle/rounded rectangle for the process, squares/rectangles for entities, labelled arrows for flows; no data-store symbol present at this level.
Total	15	—

(c) Expand on the context level data flow diagram by creating a level 0 data flow diagram.

(25 marks)

FULL MODEL ANSWER

Expanding the context diagram means "opening up" process 0 into the major internal processes that together deliver the same external behaviour already agreed in part (b) — every external entity and every external data flow from the context diagram must still appear, unchanged, at this level (the balancing rule). Reading through the scenario's three paragraphs in order gives five major internal processes:

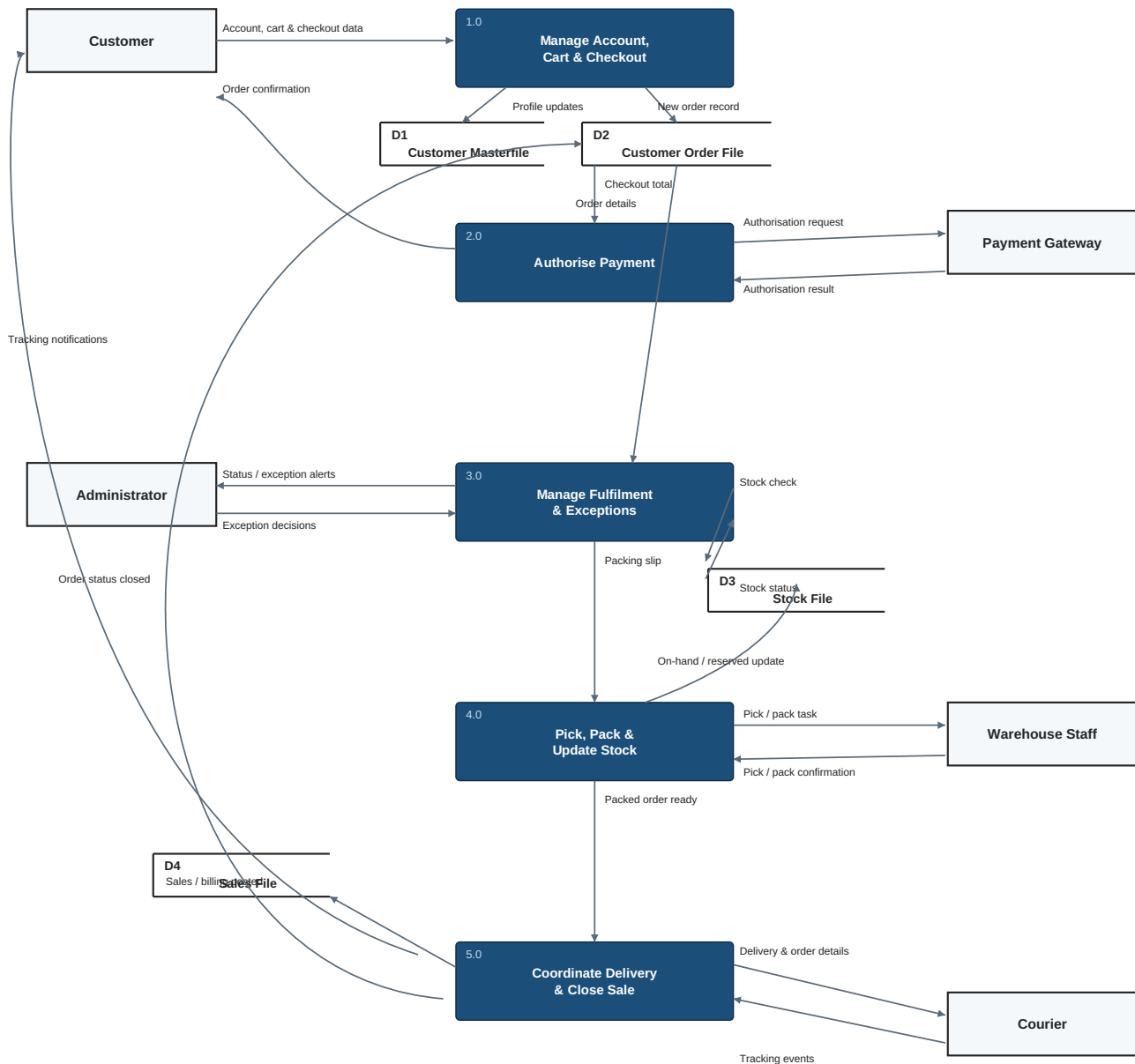


Figure 1(c). Level 0 DFD: decomposition of the Gym+Coffee order-fulfilment system into five numbered processes and four data stores.

Process	Covers (from the scenario)
1.0 Manage Account, Cart & Checkout	Create/update account; add items to cart; confirm purchase at checkout.
2.0 Authorise Payment	Submit payment; exchange authorisation request/result with the Payment Gateway; issue order ID and confirmation.
3.0 Manage Fulfilment & Exceptions	Administrator monitors progress; resolves failed authorisation, address mismatch, or out-of-stock exceptions; prepares order for picking.
4.0 Pick, Pack & Update Stock	Warehouse locates SKUs, confirms picks, packs the parcel; on-hand/reserved stock quantities updated.
5.0 Coordinate Delivery & Close Sale	Courier booked and tracked through to delivery; billing and sales figures posted once delivery is confirmed.

Four data stores hold the persistent data referenced explicitly in the scenario: **D1 Customer Masterfile** (profile changes), **D2 Customer Order File** (the order record, items, totals, timestamps, and status through its lifecycle), **D3 Stock File** (on-hand/reserved quantities), and **D4 Sales File** (billing and sales figures posted at close-out) — matching the scenario's own statement that these three files "are kept accurate and in sync" throughout.

Mark Allocation — (c)

Component	Marks	Criteria for full marks
Processes correctly identified, numbered, and decomposed from the context diagram	10	5 major processes present (2 marks each), each representing a genuine transformation of data (not a mere data movement), consistently numbered 1.0–5.0.
Data stores correctly identified and labelled	5	D1–D4 all present, each holding data genuinely described as persisted in the scenario, correctly drawn with open-ended store notation.
Data flows accurately connect processes, stores and entities	7	Flows correctly labelled and directed; no data store connected directly to another store or to an external entity without an intervening process (DFD rule).
Consistency (balancing) with the context diagram in part (b)	2	Same five external entities appear; no external flow from part (b) is dropped or contradicted.
Correct DFD notation throughout	1	Rectangles for processes, open rectangles for stores, squares for entities, labelled directional arrows.
Total	25	—

Question 1 Total

50 marks

SECTION B · OPTION A

SECTION B — ANSWER ONE QUESTION FROM THIS SECTION

Question 2

(a) Calculate the expected value of the cutover weekend outcomes in Table 1.

(10 marks)

FULL MODEL ANSWER

Wording note. Part (a) asks for "the expected value of the testing phase," but no testing-phase probabilities are given anywhere in Question 2 — Table 1 gives outcomes for the cutover weekend described in the stem. This answer computes the expected value of the cutover weekend outcomes in Table 1, which is the only data the question actually supplies, and is very likely the intended scenario given the surrounding context (production cutover, go-live risk register).

Expected value is calculated as the sum, over every possible outcome, of that outcome's probability multiplied by its value:

$$EV = \sum (\text{probability of outcome} \times \text{value of outcome})$$

Outcome	Probability	Value (€)	Probability × Value
On-time Cutover, No Data Loss	35%	+25,000	+8,750
Minor Reconciliation Issues	25%	−3,000	−750
Delayed Cutover (≤ 24 hours)	20%	−12,000	−2,400
Partial Rollback	12%	−28,000	−3,360
Full Rollback	8%	−60,000	−4,800
Expected Value	100%	—	−€2,560

The probabilities sum to 100%, confirming the five outcomes are exhaustive and mutually exclusive as stated. The expected value is **−€2,560**: on average, across many cutover weekends run under these same conditions, the project manager should expect a net financial cost of €2,560 from cutover-related risk, even though the single most likely individual outcome (on-time cutover) is actually a €25,000 gain. This is the key insight expected-value analysis surfaces: the most probable single outcome and the expected (average) outcome are not the same thing once the tail-risk outcomes (partial/full rollback) are weighted in.

Mark Allocation — (a)

Component	Marks	Criteria for full marks
Correct formula/method stated	2	States or clearly applies $EV = \Sigma(\text{probability} \times \text{value})$.
Correct substitution of all five outcomes	4	Each of the 5 probability×value products correctly calculated (0.8 marks each).
Correct final arithmetic and expected value stated	3	Products correctly summed to –€2,560, with correct sign.
Clear presentation and brief interpretation	1	Working shown in a table or equivalent structured form; sign/meaning of a negative EV briefly interpreted.
Total	10	—

(b) Using the stakeholder information in Table 1, populate a Stakeholder Power–Interest Grid. Treat **(20 marks)** "power" as the ability to materially enable/block decisions or resources; treat "interest" as the degree to which outcomes affect the stakeholder's goals/operations.

FULL MODEL ANSWER

A Power–Interest grid plots each stakeholder on two dimensions: **power** (their ability to materially enable or block project decisions or resources) on the vertical axis, and **interest** (how much project outcomes affect their own goals or operations) on the horizontal axis. This produces four quadrants, each implying a different default engagement strategy.

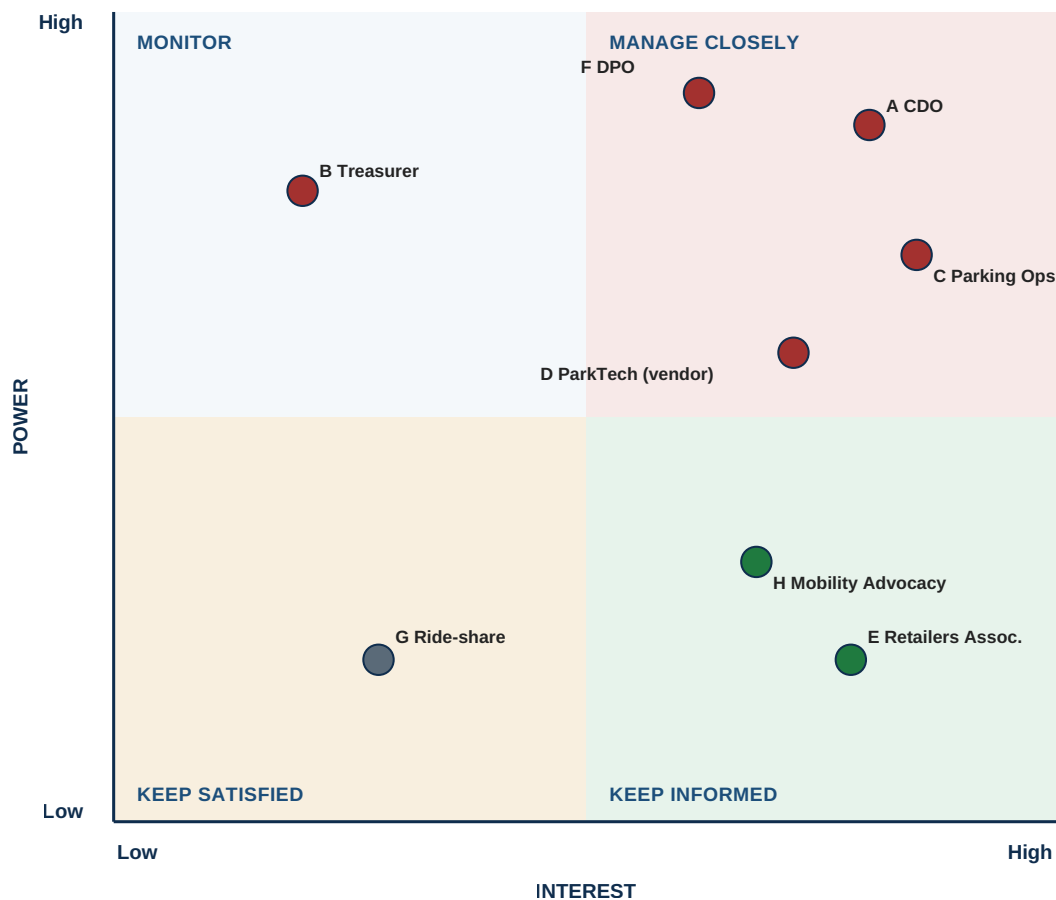


Figure 2(b). Power–Interest grid for the municipal Smart Parking & Mobility Platform, plotting the eight stakeholders from Table 1.

ID	Stakeholder	Power	Interest	Quadrant
A	City Chief Digital Officer	High — controls architecture standards and budget	High — KPIs tied to rollout success	Manage Closely
B	City Treasurer	High — releases capital/operating budgets	Low — limited involvement in features/operations	Keep Satisfied
C	Parking Operations & Enforcement	High — operational acceptance essential for rollout	High — owns day-to-day workflows directly affected	Manage Closely
D	ParkTech Ltd. (Prime Vendor)	High — holds platform IP and delivery control	High — reputation and contract tied to outcome	Manage Closely
E	Local Retailers Association	Low — no budget or decision authority	High — parking turnover directly affects footfall	Keep Informed
F	Data Protection Officer	High — can halt go-live on compliance grounds	High — DPIA/consent outcomes are their core remit	Manage Closely
G	Ride-share Platforms	Low — optional API consumers, not critical to MVP	Low — integration beneficial but not essential to them	Monitor
H	Citizens' Mobility Advocacy Group	Low — no formal decision or budget authority	High — accessibility, UX and pricing are their core mission	Keep Informed

D (ParkTech) and F (DPO) are the two genuinely arguable placements. A vendor with contractual delivery obligations is sometimes treated as "Keep Satisfied" rather than "Manage Closely" if the PM's framework defines power narrowly as internal organisational authority; here it is scored as "Manage Closely" because the scenario explicitly gives the vendor platform IP and named quality/timeline commitments, which is a material ability to affect the project's core deliverable, not just its budget. Similarly the DPO could be argued into "Keep Satisfied" if their interest were read as narrow (compliance sign-off only); it is scored "Manage Closely" here because a DPIA/consent halt is both high-power (an effective veto) and high-interest (privacy-by-design is their entire remit on this project). A student who places either stakeholder in the adjacent quadrant with a clear, scenario-grounded justification should receive full credit for that stakeholder.

Mark Allocation — (b)

Component	Marks	Criteria for full marks
Grid correctly constructed	4	Both axes present and correctly labelled (Power vertical, Interest horizontal); four quadrants clearly delineated and named.
Placement of the 8 stakeholders	16	2 marks per stakeholder: correct or well-justified quadrant, with a brief reason grounded in the stakeholder's actual description in Table 1 (not just the label alone).
Total	20	—

(c) Discuss how this tool may help project managers make strategic decisions and prioritize stakeholder engagement in the context of this project.

(20 marks)

FULL MODEL ANSWER

The Power–Interest grid gives a project manager a single, defensible way to decide *where limited engagement effort should go*, rather than treating every stakeholder as equally deserving of time and communication. In the Smart Parking & Mobility Platform, that translates into four concrete strategic decisions:

1. Concentrate effort on the high-power/high-interest cluster first. The CDO, Parking Operations, ParkTech, and the DPO all sit in "Manage Closely." Because each of these can materially block or delay go-live (budget sign-off, operational acceptance, delivery, and compliance halt respectively), the grid tells the PM these four relationships need a joint governance forum — e.g. a steering group with regular structured check-ins — rather than ad-hoc updates, because a failure with any one of them stops the project outright.

2. Avoid under-investing in "Keep Satisfied" stakeholders just because they are quiet. The City Treasurer has low day-to-day interest but high power (they release the budget). The grid's discipline is precisely to stop a PM reading "low interest" as "low priority" — the Treasurer still needs periodic, concise budget-focused updates, timed around funding decision points, even though they don't need the same operational detail as Parking Operations.

3. Use "Keep Informed" stakeholders as an early-warning and legitimacy channel, not a reporting burden. Local Retailers and the Citizens' Mobility Advocacy Group have real stakes (footfall, accessibility, pricing fairness) but no formal power to block the project. The grid recommends transparent, regular information flow to these groups — consultation sessions, published FAQs — both to surface UX or fairness problems while they are still cheap to fix, and because a currently low-power group whose concerns are ignored can escalate through local media into something that does affect delivery.

4. Apply minimal but non-zero effort to "Monitor" stakeholders, and watch for quadrant drift. Ride-share platforms need little proactive engagement now, but the grid is not static: if curb-management APIs later become central to the platform's business model, their power or interest could rise, moving them into a different quadrant and warranting a change in engagement strategy. A good PM re-plots the grid at major milestones rather than treating the initial placement as fixed for the life of the project.

Taken together, the tool turns a long, undifferentiated stakeholder list into a small number of engagement tiers, each with a distinct communication cadence and purpose — which is what allows a PM to prioritise scarce engagement time defensibly, rather than by instinct or the loudest voice in the room.

Mark Allocation — (c)

Band	Marks	Descriptor
Excellent	18–20	Clearly explains at least 3 distinct strategic uses of the grid (prioritisation, resourcing, risk of under/over-engagement, quadrant drift over time), each explicitly grounded in this project's own stakeholders — not generic PM theory alone.
Good	13–17	Explains 2–3 strategic uses with some grounding in the specific stakeholders/scenario, but with less depth or partial grounding in Table 1.
Partial	8–12	Correctly describes what the grid shows and gives at least one generic strategic implication, but with little or no connection back to this project's stakeholders.
Minimal	0–7	States that the grid helps prioritise stakeholders without explaining how or why, or largely restates part (b) without new analysis.

Question 2 Total	50 marks
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SECTION B · OPTION B

SECTION B — ANSWER ONE QUESTION FROM THIS SECTION

Question 3

(a) Draw and label the V&V model.

(10 marks)

FULL MODEL ANSWER

The Verification & Validation (V&V) model extends the waterfall SDLC into a "V" shape, pairing each specification/design stage on the descending left arm with a corresponding testing stage on the ascending right arm. The left arm answers "are we building the product right?" (verification); the right arm answers "are we building the right product?" (validation).

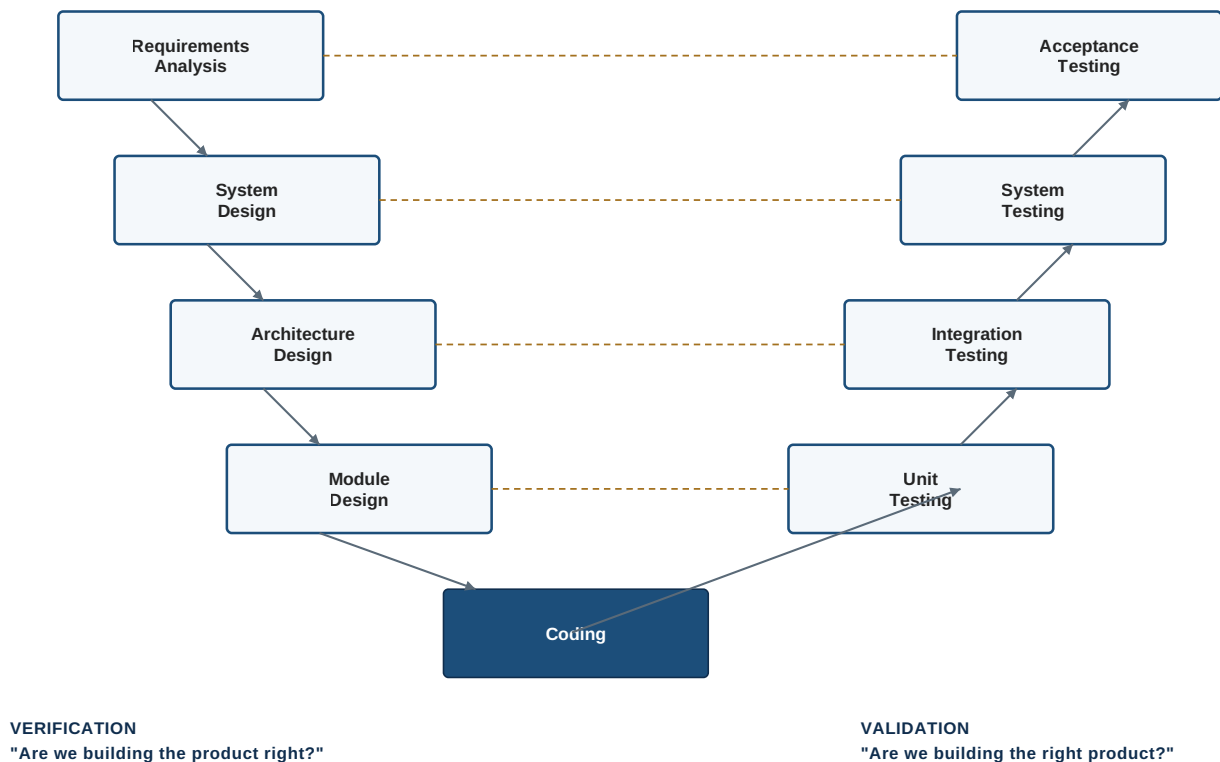


Figure 3(a). The V&V model, pairing each SDLC development stage with its corresponding test level.

Each dashed correspondence line reflects what that test level is actually checking against: unit testing verifies individual modules against the module design; integration testing verifies module interactions against the architecture design; system testing verifies the whole system against the system design; and acceptance testing validates the delivered system against the original requirements — the two ends of the "V" meeting back where the project started, from the user's point of view rather than the developer's.

Mark Allocation — (a)

Component	Marks	Criteria for full marks
Correct V-shape structure with a verification (left) and validation (right) arm	3	Two clearly distinct arms shown descending then ascending, meeting at a bottom vertex (Coding).
Four test levels correctly placed and labelled on the validation arm	4	Unit, Integration, System, and Acceptance Testing all present, in the correct ascending order (1 mark each).
Correct correspondence shown between matched verification/validation stages	2	Each test level linked to the design/requirements stage it actually verifies against (e.g. Unit–Module Design, Acceptance–Requirements).
Clear, correctly labelled diagram	1	All boxes labelled; diagram legible and unambiguous.
Total	10	—

(b) Describe the different test levels (unit, integration, system, acceptance).**(20 marks)****FULL MODEL ANSWER**

Test level	Description
Unit Testing	Tests individual methods, classes, or components in isolation, before they are integrated with the rest of the software. Its goal is to identify and fix as many defects as possible while a problem is still confined to a single, small unit — the cheapest point in the SDLC to catch and correct an error.
Integration Testing	Brings together the modules that make up a program and evaluates the behaviour of the group of components once connected, rather than each in isolation. It targets errors that unit testing structurally cannot catch — interface inconsistencies between components, parameter-value anomalies (e.g. an unexpected negative value passed between modules), and run-time exceptions caused by conflicting resource needs.
System Testing	Evaluates the overall functionality, usability, and behaviour of the entire system as one integrated whole, in an environment that closely simulates real-world conditions. It covers both functional testing (does it meet the specified requirements?) and non-functional testing (performance, usability, security, compatibility), and is typically concentrated towards the end of the project before major release milestones.
User Acceptance Testing (UAT)	The final stage of testing, in which real users interact with the system to determine whether it functions as expected and is easy to use. It is the most common form of usability testing and is normally a formal milestone requiring completion and sign-off by the client before the system goes live.

Mark Allocation — (b)

Component	Marks	Criteria for full marks
Unit Testing	5	Correctly describes scope (individual methods/classes/components), timing (before integration), and goal (early, cheap defect detection).
Integration Testing	5	Correctly describes scope (groups of connected modules) and gives at least one example category of error unique to integration (interface, parameter, or run-time).
System Testing	5	Correctly describes scope (whole integrated system), covers both functional and non-functional testing, and notes realistic test environment/timing.
Acceptance Testing	5	Correctly describes real-user involvement, functional + usability focus, and its role as a formal, client-signed-off final milestone.
Total	20	—

(c) Apply the four test levels (unit, integration, system, acceptance) to Revolut. For each of four test **(20 marks)** areas you select, map it to a FURPS category (as explored in class) and briefly describe how you would test it.

FULL MODEL ANSWER

Revolut's own scenario supplies four natural, non-overlapping test areas — one per test level — each mapped here to a distinct FURPS category (as taught in the module: **F**unctional, **O**perational/Usability, **P**erformance, **R**eliability, **S**ecurity):

Test level	FURPS category	Test area & how it would be tested
Unit	Functional	The currency-conversion function: test in isolation that, given a quoted exchange rate and an amount, the function returns the correct converted value and correctly applies any fee — run with a fixed set of known rate/amount pairs and asserted against the expected output.
Integration	Security	The payment module's link to the external identity-verification (KYC) and card-payment gateway services: test that a failed identity check correctly and consistently blocks the associated money transfer end-to-end, rather than the transfer silently succeeding because the two services disagree on the verification outcome.
System	Performance	End-to-end load testing of the full app under a simulated spike in concurrent currency-exchange activity (e.g. during sudden market volatility): measure response time, stability, and resource utilisation under that peak load against Revolut's stated performance expectations.
Acceptance	Usability	UAT with real users completing core tasks — sending money to a contact, setting a budget, checking a real-time notification — measuring task success rate, time on task, and user satisfaction to confirm the interface is intuitive and error-free before sign-off.

The five FURPS categories used above (Functional, Operational, Performance, Reliability, Security) follow the specific naming used in this module's own lecture material, rather than the more generic textbook FURPS acronym (Functionality, Usability, Reliability, Performance, Supportability). The two schemes overlap closely but are not identical (the module's "Operational" corresponds to textbook "Usability," and the module treats "Security" as its own category rather than folding it under "Supportability"); a student using the standard textbook five-letter acronym instead should not be penalised, provided the mapping to each test area is sound.

Mark Allocation — (c)

Component	Marks	Criteria for full marks
Unit Testing → Functional	5	Correct/defensible FURPS category (2) and a Revolut-specific, correctly-scoped unit-level test description (3).
Integration Testing → Security	5	Correct/defensible FURPS category (2) and a Revolut-specific, correctly-scoped integration-level test description (3).
System Testing → Performance	5	Correct/defensible FURPS category (2) and a Revolut-specific, correctly-scoped system-level test description (3).
Acceptance Testing → Usability	5	Correct/defensible FURPS category (2) and a Revolut-specific, correctly-scoped UAT-level test description (3).
Total	20	—

Question 3 Total	50 marks
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